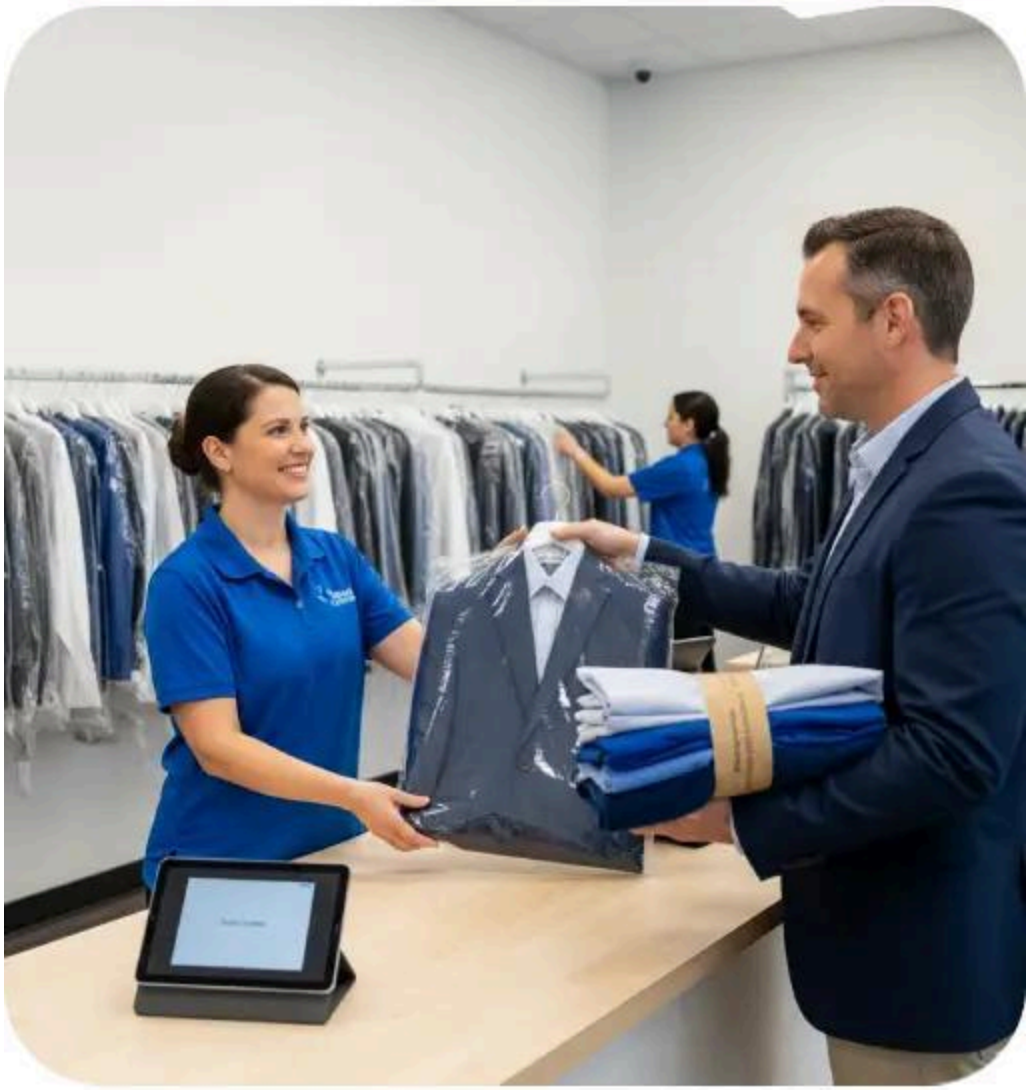
Complete Dry Cleaning Store Management Software — Full Feature & Development Blueprint

If you want to build professional dry cleaning software for a dry cleaning shop, laundry business, steam ironing shop, or multi-branch garment care company, I recommend developing a complete Dry Cleaning Management System rather than just a billing application.

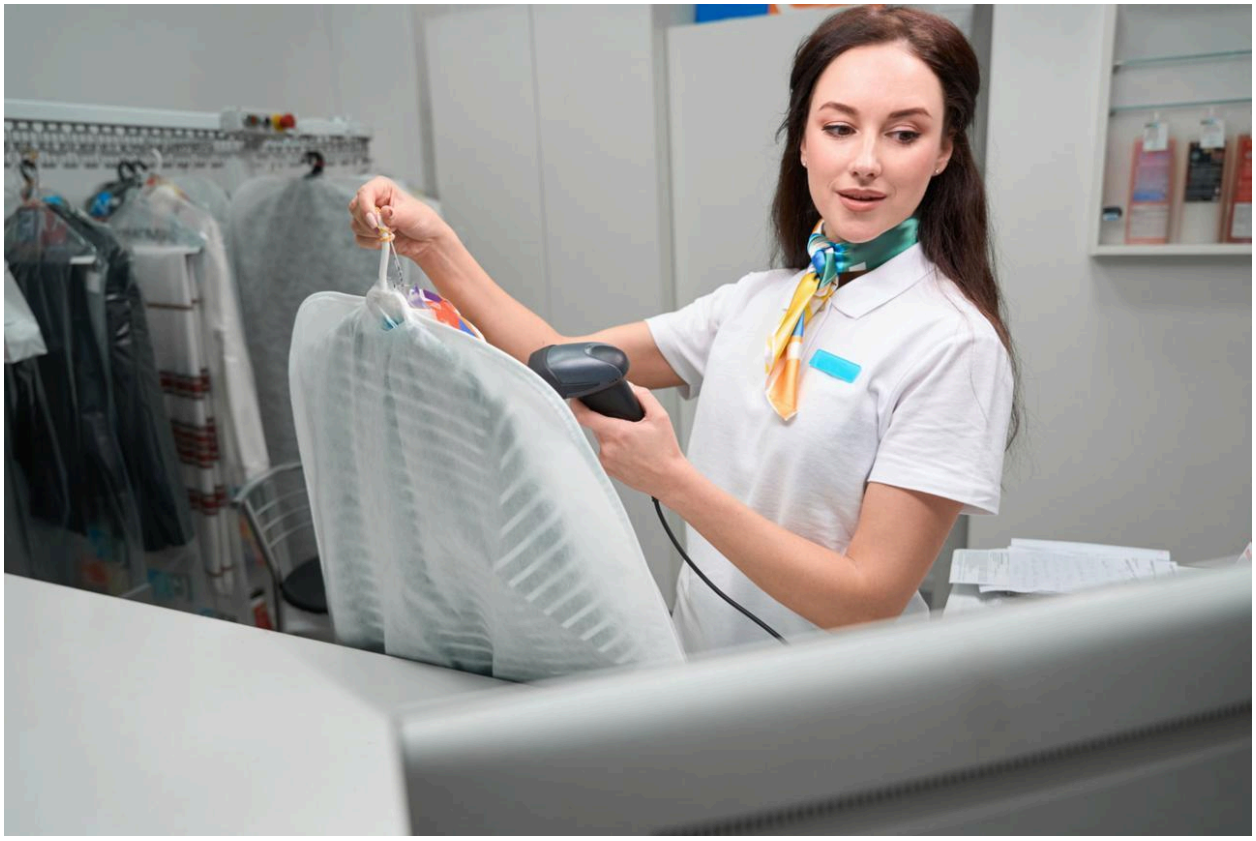
The software should manage the entire business lifecycle, from registering a customer and receiving garments to tagging, cleaning, quality checking, billing, delivery, payments, and customer feedback.

It should work for a small local dry cleaning shop as well as a commercial laundry chain with multiple outlets.



1. Customer management

Store complete customer profiles, addresses, phone numbers, order history, garment preferences, outstanding balances, and complaints.



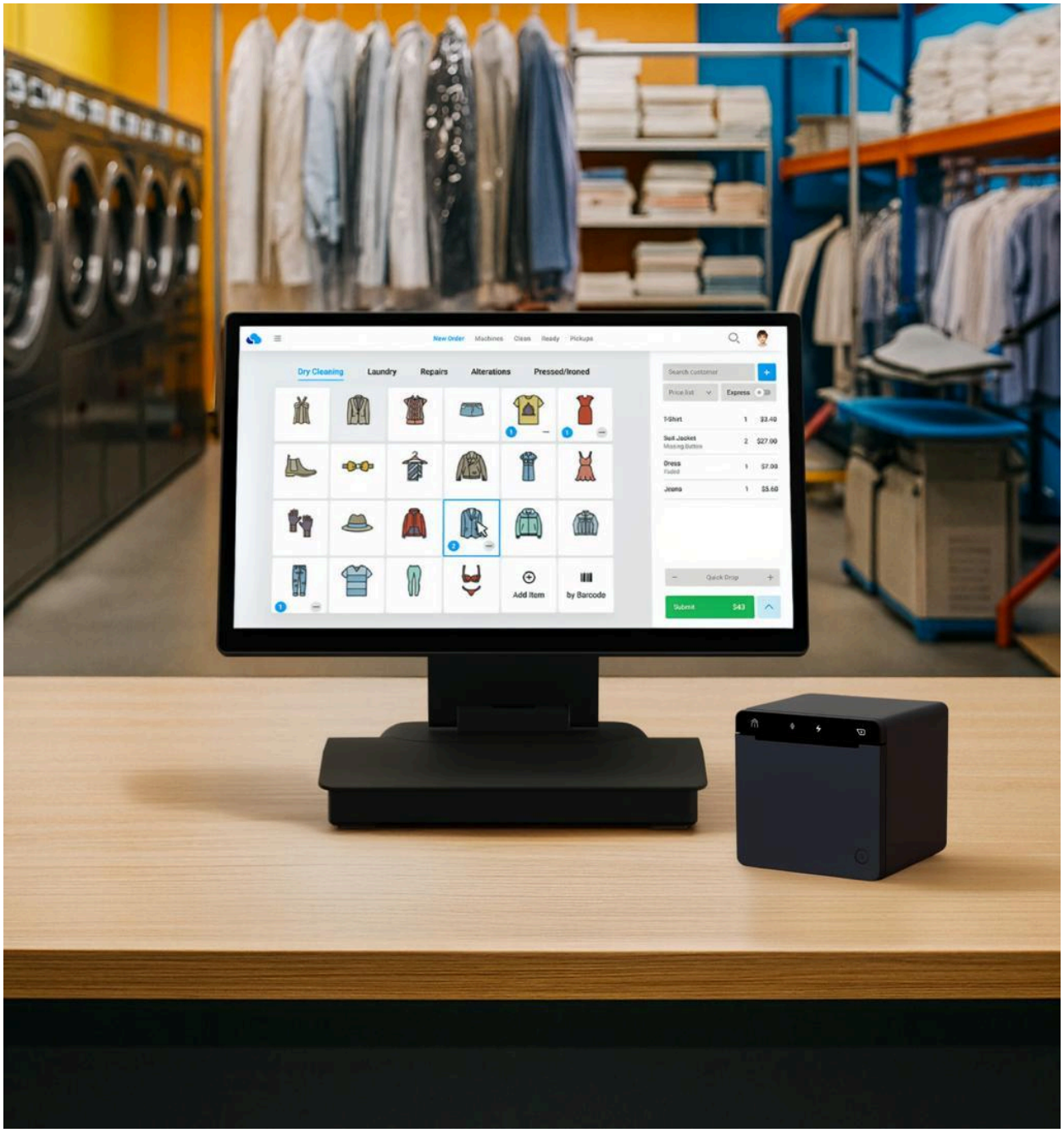
2. Order and garment tracking

Create job orders, record each garment, print barcode tags, track cleaning stages, and prevent garments from being misplaced.



3. Operations and delivery

Manage cleaning, stain treatment, pressing, quality control, packaging, pickup, home delivery, and order status notifications.



4. Billing and business management

Generate invoices, accept payments, manage expenses, track staff performance, calculate revenue, and review business reports.

1. Define the software structure

Build the system as five connected parts:

Store POS / Reception Panel

Customer registration, garment intake, orders, billing and delivery

Backend API and Business Logic

Orders, pricing, permissions, payments, notifications and workflows

Database

Customers, garments, orders, payments and stock

Notifications

SMS, WhatsApp, email and in-app alerts

Owner / Admin Panel

Reports, staff, pricing, branches, expenses, configuration and audit logs

The software should support desktop computers, tablets and mobile browsers. You can add a customer-facing app or online booking portal later without rebuilding the core system.

2. Complete customer management module

This is one of the most important modules because the store must know exactly who the customer is, what they submitted, what they paid, and what is still pending.

Customer 360° Profile

One customer record containing all interactions and transactions.

A. Personal details

- Unique customer ID and customer code
- Full name, mobile number, alternate phone number
- Email address, date of birth and optional anniversary
- Complete address, locality, city, state and PIN code
- Customer photo, if needed and consented to
- Customer type: walk-in, regular, VIP, corporate or online
- Registration date, referred by and assigned branch

B. Service preferences

- Preferred cleaning service and detergent preferences
- Special fabric-care instructions
- Preferred pickup and delivery times

- Preferred language and communication channel
- Allergies or fabric-care sensitivities, where voluntarily provided and relevant

C. Customer account

- Total orders, total spending and average order value
- Last order date and most frequently used services
- Pending orders and garments ready for collection
- Advance balance, store credit and loyalty points
- Unpaid invoices and payment history
- Complaints, refunds, service ratings and internal notes

Customer management actions

Every customer record should have these actions:

- Add customer, edit details, search, filter and merge duplicate records.
- View complete order and payment history.
- Create a new order directly from the profile.
- Send payment reminders and ready-for-pickup messages.
- Add a customer to a loyalty or membership plan.
- Download customer statements and export permitted records.
- Mark a customer inactive without deleting transaction history.

Important: Mobile number should be a primary search option, but the system must support customers who do not have one. Keep personal data access restricted by staff role.

3. Order booking and garment intake

When a customer arrives with clothes, the counter employee should be able to create an order quickly without opening multiple screens.

Example: New dry cleaning order

Order ID

DC-2026-00125

Customer

Rahul Sharma

Received

22 Oct 2026

Expected delivery

24 Oct 2026

Garment list



Suit jacket

Navy blue · Dry clean · 1 pc

Barcode: G-00125-01

₹250



Formal shirt

White · Steam iron · 2 pcs

Individual garment tags

₹160

Subtotal

₹410

Advance paid

₹200

Balance due

₹210

Illustrative order and prices, not a live transaction.

Required order fields

Field	Purpose
Order number	Unique, sequential ticket
Customer	Link to an existing or new customer
Branch	Store receiving the garments
Order source	Walk-in, phone, website or pickup
Garment list	Every item and its quantity
Service type	Dry clean, wash, iron, steam, stain treatment, etc.

Garment condition	Existing stains, tears, missing buttons, colour fading
Garment details	Type, colour, brand, fabric and optional photos
Instructions	Customer's special requests
Due date	Promised completion date and time
Charges	Service price, extras, discounts and applicable taxes
Payment	Advance, partial payment, full payment or credit
Delivery method	Counter collection or home delivery

Essential garment-level features

Each garment should have its own record, even if multiple garments belong to one order.

- Unique barcode or QR code.
- Garment type, colour, brand and fabric.
- Service selected and garment-specific price.
- Intake condition, photographs and damage notes.
- Stain locations, treatment instructions and care warnings.
- Current processing stage and physical storage location.
- Re-cleaning, alteration or repair requirements.
- Final quality-control result and delivery confirmation.

This garment-level tracking is a core feature in established dry cleaning software, alongside order intake, barcode labels, processing and customer notifications.



Laundry POS| Dry Cleaning POS
+1

4. Complete garment processing workflow

The software must show where every order and garment is at any moment.

01

Received at counter

Record garment details, condition and promised delivery.

02

Inspection and tagging

Assign barcode, identify stains and photograph existing damage.

03

Sorting

Group by fabric, colour, service type and treatment requirements.

04

Cleaning / treatment

Track dry cleaning, washing, stain removal and special care.

05

Drying and finishing

Track drying, steam ironing, pressing and alterations.

06

Quality control

Inspect cleanliness, stains, damage, buttons and finishing.

07

Rework if required

Return failed items to the appropriate processing stage.

08

Packing and storage

Group garments, scan tags and assign rack or shelf locations.

09

Ready for collection

Notify the customer and show the amount still due.

10

Delivered / closed

Verify collection, record payment and issue a receipt.

Rules that make tracking reliable

1. Staff scan the garment barcode when moving an item to a new stage.
2. Record the employee, timestamp, previous stage and new stage.
3. Prevent delivery until required quality checks are passed, except for an authorized override.
4. Alert managers when an order is overdue or a garment is missing from its expected location.
5. Support partial orders, such as five garments ready while one is still being processed.
6. Support cancellations, rejected garments, re-cleaning, damage claims and refunds.
7. Allow configurable workflows for items sent to an external cleaning plant.

Do not let an employee simply change a garment to “Delivered” without recording the actual delivery event and any outstanding balance.

5. Billing, payments and invoices

The POS should calculate all charges automatically using the configured price list.

Payment modes for an Indian dry cleaning business

- Cash

- UPI, including supported QR payment workflows
- Debit and credit cards through a payment terminal or gateway
- Bank transfer
- Advance payment and partial payment
- Customer credit or approved corporate billing
- Store wallet or membership credit, if enabled

Billing functionality

- Item-wise and weight-based billing.
- Service-specific price lists and additional treatment charges.
- Express or same-day service surcharge.
- Coupons, discounts and manager-authorized price overrides.
- Configurable taxes and GST invoice settings, where applicable.
- Advance adjustment, split payments and outstanding dues.
- Refunds, credit notes and cancellation adjustments.
- Printable invoice, payment receipt and customer statement.
- Daily cash register and shift closing reconciliation.
- Automatic payment reminders for overdue accounts.

The system should retain a payment ledger rather than simply overwriting an order's payment status. Every collection, refund and adjustment must be traceable.

For GST compliance, confirm the applicable tax treatment, SAC classification, invoice fields and reporting requirements with a qualified Indian accountant before launch.

6. Pickup and home delivery management



Pickup scheduling

Record customer address, preferred date, time slot, pickup notes and assigned driver. Support recurring pickup requests and express collections.



Dispatch and delivery

Assign orders to a driver, group nearby stops, verify garment bags and capture delivery confirmation.



Proof of delivery

Capture OTP verification, customer signature or an authorized delivery photo, with a timestamp.

The driver interface should show only the assigned route, customer contact details needed for the delivery, order reference and delivery instructions. It should not expose the complete customer database or financial reports.

7. Notifications and customer communication

Configure automated messages for these events:

Trigger	Customer message
Order booked	Order number, garment count and promised date
Pickup scheduled	Date, time slot and driver details if appropriate

Processing update	Current order status, if enabled
Ready for collection	Ready status, collection hours and outstanding balance
Delivery dispatched	Delivery window and tracking details
Payment received	Amount, payment mode and receipt
Payment overdue	Due amount and payment instructions
Order delayed	Updated expected completion date
Feedback requested	Rating or complaint link after delivery

Support SMS, WhatsApp Business Platform and email through approved providers. Templates should be configurable, and marketing messages should respect customer consent and opt-outs.

8. Complete owner and admin panel

The owner needs a separate dashboard that shows the overall business performance, while counter staff need a simpler operational interface.

Admin dashboard overview

Today's revenue

₹24,850

New orders

38

Ready garments

64

Overdue orders

4

Illustrative dashboard figures only.

Admin modules

Module	Functionality
Dashboard	Revenue, orders, pending work, overdue jobs and daily collections
Customer management	Profiles, history, preferences, complaints and customer balances
Order management	Search, create, edit, cancel and track orders
Garment catalogue	Garment types, fabrics, colours, brands and care instructions
Services and pricing	Rates, add-ons, taxes, discounts and express fees

Production	Processing queue, stage scanning, quality control and rework
Inventory	Detergents, solvents, packaging, hangers, tags and stock adjustments
Staff and HR	Employees, roles, shifts, attendance and optional payroll
Expenses	Rent, utilities, salaries, supplies, maintenance and other expenses
Payments	Collections, refunds, advances, dues and reconciliation
Marketing and loyalty	Memberships, coupons, customer segments and consent-based campaigns
Pickup and delivery	Drivers, assignments, routes, delivery confirmation and failed attempts
Reports	Revenue, orders, service mix, outstanding balances and profitability
Branch management	Multiple outlets, transfers, central pricing and branch-wise reports
Settings	Invoice formats, sequence numbers, notifications, taxes and business details
Security and audit	Roles, permissions, login history, activity logs and backups

Comparable commercial systems also offer inventory, customer retention, shift closing, multi-store operations, plant management and audit reporting.



Laundry POS| Dry Cleaning POS
+1

9. Staff roles and permissions

Every employee should log in using their own account. Avoid a single shared login for the whole store.

Role	Access
Super Admin	All branches, system settings, users and configuration
Owner	Business-wide reports, revenue, expenses and customer data
Store Manager	Branch operations, staff, approvals, pricing within permissions
Reception / Counter Staff	Customer registration, order intake, invoices and collections
Cleaning Staff	Assigned garments, cleaning instructions and processing updates
Quality Inspector	Inspection, rework and quality approval
Packing Staff	Packing lists, rack locations and ready orders
Cashier	Payments, receipts, refunds if authorized and shift closing

Delivery Agent	Assigned pickups, deliveries and proof of handover
Accountant	Financial reports, expenses, taxes and reconciliation
Customer	Own profile, orders, invoices and online payments

Use permissions at the action level. For example, a cashier might accept payments but not issue refunds, and counter staff might create a discount request but not approve unlimited discounts.

10. Inventory and consumable management

Dry cleaning stores also consume materials and need to track supplies.

- Cleaning solvents and detergents.
- Stain-removal products and finishing chemicals.
- Hangers, garment covers, packaging bags and tags.
- Buttons, zippers and alteration materials.
- Minimum stock levels and low-stock alerts.
- Supplier details, purchase orders and purchase invoices.
- Stock received, consumed, wasted, returned and adjusted.
- Expiry dates and batch tracking for applicable chemicals.
- Supplier payments and purchase history.

For chemical handling, maintain relevant safety information and access restrictions. Do not treat inventory software as a substitute for safe chemical storage and handling procedures.

An advanced version can deduct estimated materials from stock based on service type, then reconcile estimated consumption against actual use.

11. Reports and business analytics

The system should provide searchable reports with date, branch, service, employee and payment-mode filters, plus Excel and PDF exports where appropriate.

Essential reports

Sales and revenue

Daily, weekly, monthly and annual sales; discounts, taxes, refunds and net collections.

Customer reports

New and repeat customers, top customers, inactive customers and retention.

Order reports

Orders received, completed, pending, cancelled, delayed and delivered.

Garment reports

Garments by service, processing stage, missing item, damage claim and rework.

Outstanding dues

Customer balances, ageing of receivables, advance payments and overdue invoices.

Cash and payment reconciliation

Cash, UPI, cards, bank transfers, refunds and cashier shift differences.

Staff performance

Orders processed, stage turnaround, quality failures and deliveries completed.

Expense and profit reports

Expenses by category, branch contribution, gross margin and estimated operating profit.

Inventory reports

Stock on hand, consumption, purchases, wastage and low-stock items.

Pickup and delivery

Scheduled, completed, delayed, failed and rescheduled pickups or deliveries.

Important financial distinction: revenue, cash collected, outstanding receivables and profit are different figures. The dashboard must not label all collections as profit. Calculate profitability using properly defined revenue, cost and expense rules.

12. Advanced features for a commercial-grade system

These can distinguish the software from a basic billing tool.

- Garment photo inspection: Record pre-existing stains, tears and fabric damage at intake.

- QR-based customer tracking: Customers scan a code on the ticket to check permitted order information.
- Customer portal: View order history, book pickups, pay bills and submit complaints.
- Memberships and subscriptions: Monthly laundry plans, prepaid credits, family accounts and usage limits.
- Corporate accounts: Hotels, salons, restaurants, hospitals and businesses with monthly consolidated invoices.
- Multi-branch operations: Shared customer profiles, branch transfers, central price lists and consolidated reporting.
- External plant management: Send garments to a third-party cleaning facility and track dispatch and return quantities.
- Alterations and repairs: Record hemming, button replacement, stitching, repair charges and delivery dependencies.
- Lost or damaged garment claims: Document evidence, investigate, approve compensation and record resolution.
- Loyalty and referrals: Configurable points, rewards, referral credits and promotional campaigns.
- AI-assisted intake: Optional assistance identifying garments or flagging visible damage from photographs, with staff verification.
- Offline resilience: Allow selected counter workflows to queue temporarily during network failure and synchronize safely when connectivity returns.

Build the operational tracking and financial controls first. AI, predictive analytics and advanced route optimization can follow once the core data is reliable.

13. Database design — essential tables

If you are building this as custom software, plan the database around customers, orders, individual garments, workflow events and financial transactions.

Table	Main data
`organizations`	Business account, name, settings and subscription
`branches`	Store address, contact details, timezone and branch configuration

`users`	Login, staff profile, account status and assigned branch
`roles` / `permissions`	Role definitions and action-level permissions
`customers`	Customer code, name, contacts, preferences and addresses
`customer_addresses`	Multiple pickup and delivery addresses
`garment_types`	Shirt, trousers, suit, saree, blanket and other types
`services`	Dry clean, wash, iron, stain treatment and other services
`price_lists` / `service_prices`	Garment-specific rates, branch and effective dates
`orders`	Customer, branch, dates, status and totals
`order_items`	Individual garment, quantity, service, condition and price
`garment_tags`	Unique barcode/QR identifier for each physical garment
`order_status_history`	Status transitions, employee, timestamp and notes
`quality_inspections`	Inspection results, defects, photographs and rework

`payments`	Amount, mode, reference, status and transaction date
`payment_allocations`	Allocation of payments to invoices or orders
`invoices` / `refunds`	Invoice details, tax breakdown and refunds
`expenses`	Category, amount, branch and supporting document
`inventory_items` / `stock_movements`	Materials, suppliers, quantities and stock changes
`pickup_deliveries`	Address, schedule, driver and completion evidence
`notifications`	Channel, template, delivery status and retry history
`complaints` / `claims`	Customer issue, investigation and resolution
`audit_logs`	Who performed each important action and when

Database rules that matter

- Use UUIDs or another reliable internal identifier; never depend only on a human-readable order number.
- Enforce unique garment tags, order references and payment transaction references where appropriate.
- Store monetary amounts as exact decimal or integer minor-unit values, not floating-point numbers.
- Preserve order prices at the time of booking so future price changes do not alter old invoices.

- Keep a ledger for payments and refunds; use controlled corrections instead of silently overwriting financial history.
- Store timestamps consistently and display them in the relevant branch's timezone.
- Separate organization and branch data so users cannot access another business's records.
- Use database transactions for critical operations such as order creation, payment allocation and stock adjustments.
- Maintain indexes for customer phone searches, order numbers, barcode scans, branch IDs and order statuses.

14. Recommended technology stack

For a custom web-based system, I recommend this architecture:

Frontend: Next.js + TypeScript

Responsive POS, owner dashboard, customer portal and driver interface. Use Tailwind CSS and shadcn/ui.

Backend: Node.js + Fastify

REST APIs, business rules, role permissions, order workflows, payment processing and notifications.

Database: PostgreSQL + Drizzle ORM

Relational data, transaction integrity, reporting and migrations.

Redis + background jobs

Caching, notification queues, retry handling and scheduled reminders.

Hardware and integrations

Barcode scanner, thermal receipt printer, garment label printer, UPI/payment gateway and messaging provider.

Use a modular monolith for the first release rather than starting with many microservices. Keep customer management, orders, inventory, payments, notifications and reporting in well-separated backend modules.

15. Step-by-step development roadmap

Here is the order I would follow to build a reliable, production-ready application.

Phase 1

Foundation and business configuration

Create project structure, authentication and organization setup.

Create branches, users, roles and permissions.

Configure services, garment types, prices and invoice settings.

Set up PostgreSQL, Drizzle migrations, validation and audit logging.

Phase 2

Customers and POS

Build customer registration, search and profile management.

Build order creation with garment-level details and condition notes.

Implement price calculation, discounts, advances and outstanding balances.

Generate order numbers, invoices and payment receipts.

Phase 3

Garment tracking and operations

Generate unique barcode/QR tags for individual garments.

Build the processing board and scan-to-update workflow.

Implement inspection, rework, packing and storage locations.

Add order history and overdue-order alerts.

Phase 4

Payments and communications

Add cash, UPI and payment gateway integrations.

Implement payment allocation, refunds and reconciliation.

Integrate SMS, WhatsApp and email notifications.

Build notification templates, retries and delivery logs.

Phase 5

Business management

Build inventory, suppliers, purchases and expenses.

Add staff performance, shift closing and reports.

Implement pickup scheduling, driver assignments and delivery proof.

Add customer complaints, claims, loyalty and membership support.

Phase 6

Commercial release

Add multi-branch support and customer self-service portal.

Test barcode printers, thermal printers and real counter workflows.

Perform security, permissions, concurrency and backup-restore tests.

Deploy with monitoring, automated backups, staging and production environments.

16. Security, backup and production requirements

Before allowing a real store to use the system:

- Use secure password hashing, session management and optional two-factor authentication.
- Enforce permissions on the backend, not just by hiding frontend buttons.
- Validate inputs and protect against SQL injection, cross-site scripting and cross-site request forgery where applicable.
- Encrypt traffic using HTTPS and protect payment credentials and API secrets.
- Maintain immutable or tamper-evident audit records for sensitive financial and administrative actions.
- Back up the database automatically, keep off-site copies and regularly test restoration.
- Use idempotency keys for payment operations and notification retry logic to prevent accidental duplicates.
- Record consent, communication preferences and data retention rules.
- Restrict access to garment photographs, customer addresses and sensitive records.
- Establish appropriate privacy notices and compliance procedures under applicable Indian data-protection requirements.

17. Final testing checklist

Before launch, test the following real-world scenarios:

Go-live checklist

0/16

Customer and order

Register a new customer and find them by mobile number.
Create an order with several different garments and services.
Record an existing stain and photograph the garment condition.
Print individual garment tags and scan them successfully.

Processing and delivery

Move garments through every configured stage.
Prevent unauthorized delivery before quality approval.
Handle a partially ready order and a re-cleaning request.
Confirm delivery, record the final payment and print a receipt.

Financial controls

Accept advance and split payments without duplicate collection.
Change future prices without changing old invoices.
Process an authorized refund and reconcile the cash register.
Verify that revenue, collections, dues and expenses are reported separately.

Security and reliability

Confirm counter staff cannot access owner-only reports.
Confirm one business cannot access another business's records.
Restore a backup in a test environment.
Verify that a temporary network failure does not silently lose or duplicate transactions.
Copy checklist

18. The right scope for your first release

Do not try to build every advanced feature at once.

Priority	What to deliver

P0 — Must have	Login, staff permissions, customer profiles, garment intake, barcode tags, order tracking, pricing, invoices and payments
P1 — Essential operations	Quality control, packing, delivery, notifications, outstanding dues and daily reports
P2 — Business expansion	Inventory, expenses, loyalty, subscriptions, customer portal and multiple branches
P3 — Advanced	AI inspection, route optimization, offline synchronization, corporate integrations and predictive analytics

My recommendation: First build a complete single-store version that can be used at an actual counter, then make it multi-branch and subscription-ready. If you intend to sell the software to many independent dry cleaning businesses, make it multi-tenant from the beginning so each business has isolated customers, orders, staff, pricing and reports.

For reference, you can also review the feature coverage of SMART Laundry Software, Quick Dry Cleaning Software and LaundryBook. These provide useful examples of garment tracking, POS, customer records, billing and branch operations.



SMART Laundry by Shreay Technology
+2

One important decision before turning this blueprint into a development specification: are you building this software for your own dry cleaning shop, or do you want to sell it to multiple dry cleaning shops as a SaaS product? That decision changes the database architecture, subscription billing, tenant isolation and admin panel design.

****

Case Study: Complete Dry Cleaning & Laundry Management Software

Product strategy • Business analysis • System requirements • Technical architecture • Implementation plan

Project name: DryCare — Dry Cleaning Store Management System

Document type: Comprehensive Software Product Case Study

Target market: Dry cleaning shops, laundry stores, garment-care businesses, commercial laundry plants and multi-branch operators in India.

1. Executive summary

DryCare is a proposed end-to-end dry cleaning management platform designed to digitize the complete operational lifecycle of a garment-care business. It connects customer management, garment intake, barcode identification, cleaning workflows, quality control, billing, payments, pickup and delivery, inventory, staff management and business reporting in one centralized system.

Many small and medium-sized stores rely on paper receipts, registers, spreadsheets, phone calls and manual garment identification. These processes can make it difficult to find customer records, trace individual garments, identify delayed orders, reconcile payments and measure profitability.

The proposed solution replaces fragmented workflows with a unified, auditable system that records each customer interaction, garment movement and financial transaction.

Product vision

To give every dry cleaning business a reliable digital system for knowing:

Who is the customer?

Profile, preferences, history and outstanding balance.

Where is each garment?

Unique identity, current stage and physical location.

What is the financial position?

Revenue, collections, expenses, refunds and dues.

How is the business performing?

Turnaround time, service mix, productivity and margins.

This is a proposed product case study, not a report of a software deployment or measured customer outcomes. All example businesses, financial figures, targets and scenarios below are illustrative unless explicitly identified as requirements.

2. Business background

Consider a hypothetical dry cleaning company called CleanCare Garment Services, operating one main store and planning to expand to additional branches.

The business accepts suits, shirts, trousers, dresses, sarees, blankets, curtains and other garments. Services include dry cleaning, washing, steam ironing, stain removal, alterations and scheduled home pickup and delivery.

Initially, the store manages customers and orders manually. As daily volume increases, the owner faces operational problems:

- Employees cannot immediately find the details of an old order.
- Garments are identified using handwritten slips that can become detached.
- Staff have limited visibility into cleaning stages and expected completion dates.
- Customers repeatedly call to ask whether garments are ready.
- Advance payments, partial payments and credit balances are difficult to reconcile.
- Managers cannot easily distinguish busy periods, profitable services and recurring customer behavior.
- The owner cannot reliably compare branch performance or employee productivity.

The business therefore needs more than a digital receipt generator. It needs a complete operating system for garment care.

3. Problem statement and business impact

Business problem	Root cause	Proposed solution
Customer records are scattered	Paper registers and duplicate entries	Central customer database with search and history

Garments go missing or are difficult to locate	No unique item-level identity	Barcode or QR tags and location tracking
Orders are delayed	Manual scheduling and no stage visibility	Workflow queues, due dates and overdue alerts
Customers keep calling	Status information is not readily available	Automated notifications and customer tracking
Payment disputes occur	Incomplete transaction records	Payment ledger, receipts and audit trail
Stock runs out unexpectedly	No material consumption tracking	Inventory records and reorder alerts
Owner cannot measure profitability	Disconnected sales and expense data	Financial reports and service-level costing
Multiple stores operate differently	Separate registers and processes	Central configuration and branch-level controls

The core business hypothesis is that better traceability, fewer manual steps and faster access to reliable information can improve operational consistency and customer experience. The extent of any improvement must be established through a measured pilot.

4. Project objectives

The project should achieve the following measurable objectives.

01

Centralize customer information

Maintain a single customer record with contact information, addresses, order history, preferences, complaints and payment history.

02

Track every garment

Identify each physical garment with a unique barcode or QR code and record its processing stage and location.

03

Standardize operations

Define consistent intake, inspection, cleaning, quality control, packing and delivery workflows.

04

Improve financial control

Track charges, discounts, taxes, advances, payments, refunds, outstanding balances and expenses.

05

Improve customer communication

Provide timely order confirmations, ready-for-collection alerts and delivery updates.

06

Enable business growth

Support additional branches, corporate accounts, memberships and eventually a multi-tenant SaaS business.

5. Stakeholder analysis

The system must satisfy the needs of everyone involved in the garment-care lifecycle.

Stakeholder	Primary needs

Business owner	Revenue, expenses, margins, customer retention and branch performance
Store manager	Daily workload, delayed orders, staff assignments and approvals
Reception employee	Fast customer search, order intake, pricing and receipt printing
Cleaning operator	Garment instructions, processing queue and stage updates
Quality inspector	Inspection checklist, defect documentation and rework control
Packing employee	Correct garment grouping, rack location and dispatch verification
Cashier	Accurate collections, refunds, receipts and shift reconciliation
Delivery agent	Assigned deliveries, route information and proof of handover
Accountant	Financial ledger, tax records, expenses and reconciliations
Customer	Clear prices, reliable delivery, order updates and easy complaint resolution
SaaS platform administrator	Tenant onboarding, plans, support, billing and platform health

Each role should have its own permitted actions. For example, the cleaning operator can update processing status but should not be able to alter invoice amounts or access unrelated customer financial records.

6. Proposed solution and functional modules

The product consists of interconnected modules rather than independent applications.

Functional system map

Customer and order intake

Customer profiles • Garment registration • Pricing • Order ticket

Garment tracking

Tags, locations and history

Production

Sorting, cleaning, finishing and QC

Billing and payments

Invoices, collections and refunds

Fulfillment

Packing, pickup and delivery

Central business management

Reports • Inventory • Staff • Expenses • Audit logs • Branches

Module specifications

A. Customer Relationship Management (CRM)

- Customer ID, contact information and multiple addresses.
- Customer search by name, phone, ID or order number.
- Order, invoice, payment and complaint history.
- Service preferences and special garment-care instructions.
- Customer segmentation, loyalty and membership eligibility.
- Duplicate detection, privacy controls and consent management.

B. Order and garment management

- Create, edit, cancel and search orders according to permissions.
- Add multiple garment types and services to an order.
- Store individual garment condition, photos, notes and price.

- Generate garment labels and a consolidated order receipt.
- Record promised completion time and priority.
- Support partial completion, missing garments, rework and external processing.

C. Cleaning and production

- Configurable stages for each service type.
- Queue views for received, sorting, cleaning, finishing and inspection.
- Barcode scanning at designated workstations.
- Employee assignment and timestamps.
- Fabric-care warnings and stain-treatment instructions.
- Quality approval, rejection, rework and final release.
- Overdue alerts and work-in-progress reports.

D. Billing and accounting controls

- Configurable price lists and additional charges.
- Discounts, applicable taxes, deposits and partial payments.
- Cash, UPI, card, bank transfer and supported gateway payments.
- Invoice generation, receipts, refunds and credit notes.
- Customer dues, approved credit limits and payment reminders.
- Cashier shifts and reconciliation.
- Revenue, expenses, receivables and margin reporting.

E. Inventory and purchasing

- Materials, packaging, tags, hangers and other consumables.
- Suppliers, purchase orders and purchase records.
- Stock movements, wastage and minimum-stock thresholds.
- Optional consumption rules for different services.
- Supplier balances and stock valuation.

F. Delivery and fulfillment

- Pickup bookings, address validation and time slots.
- Driver assignments and dispatch manifests.
- Garment bag verification before dispatch.
- Delivery confirmation and failed-delivery handling.
- Optional route optimization and customer self-service tracking.

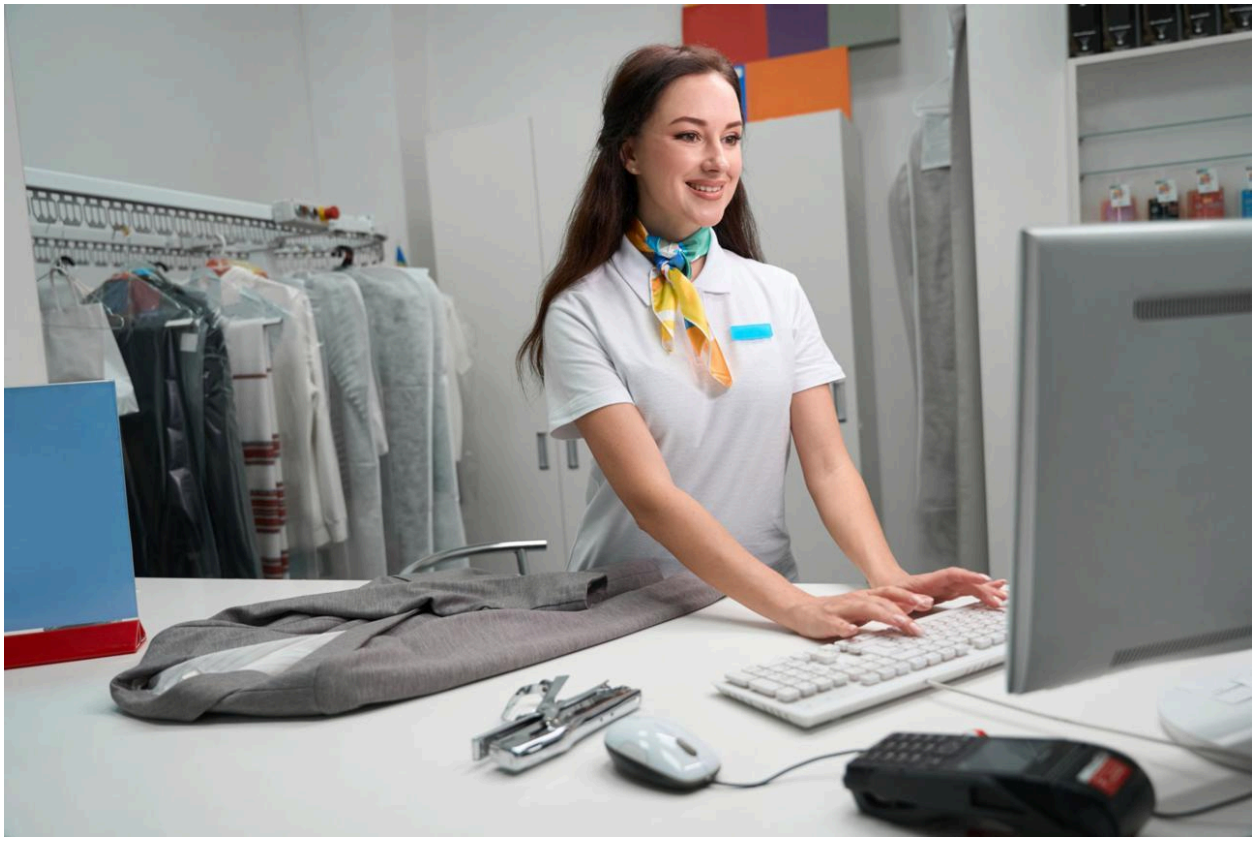
G. Reporting and analytics

- Daily and monthly sales.
- Order volume and turnaround time.
- Service-level sales and contribution margins.
- Repeat customer rate and customer retention.

- Outstanding balances and collection ageing.
- Inventory consumption and waste.
- Staff productivity, quality failures and rework.
- Branch comparisons and operating expenses.

7. End-to-end operational case scenario

The following scenario demonstrates how the proposed system should operate in a real shop.



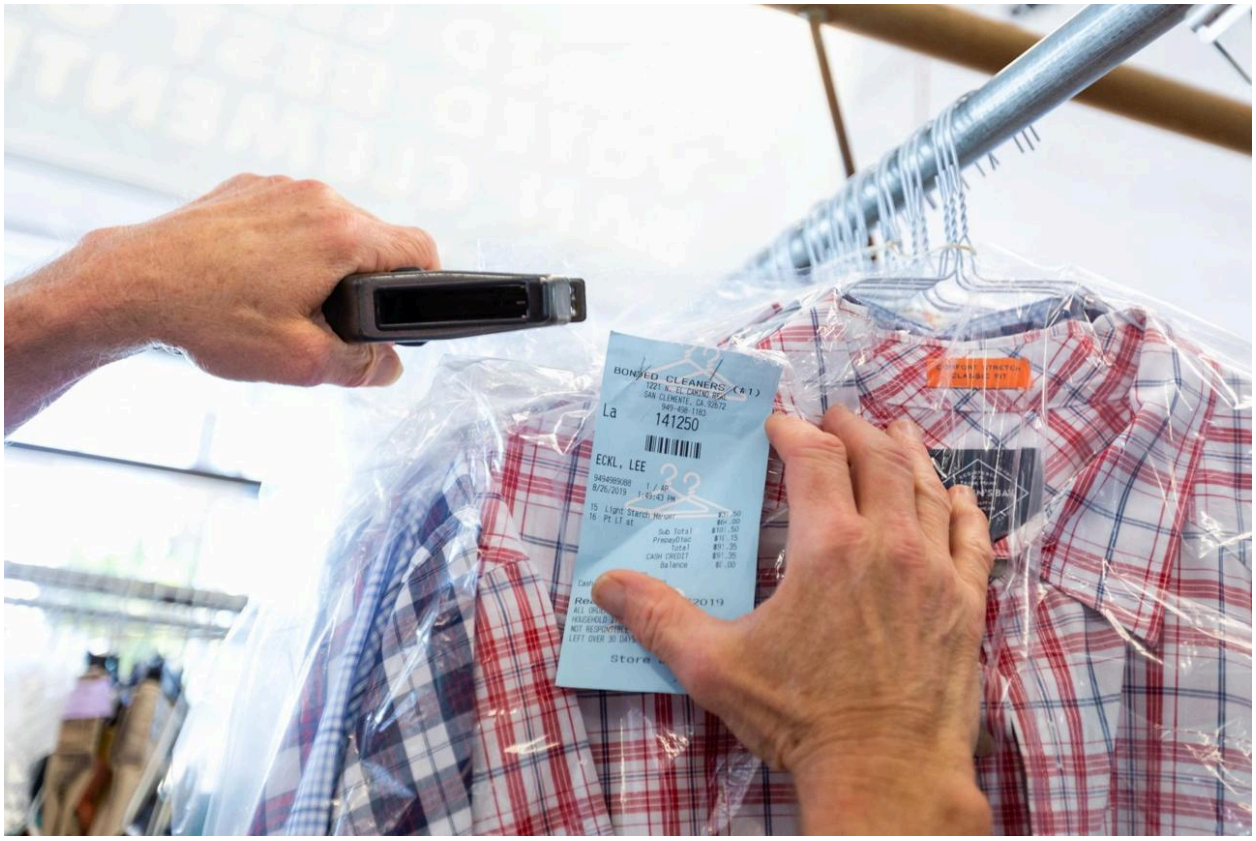
Step 1 — Customer registration

The receptionist searches the customer's mobile number. If the customer already exists, the existing profile is opened; otherwise, a new customer ID is generated.



Step 2 — Garment intake

The employee records a suit, two shirts and a pair of trousers, selects the required services and records any existing damage or stains.



Step 3 — Order confirmation

The system calculates the price, generates the order number and individual garment IDs, prints tags and records an advance payment.



Step 4 — Production tracking

Staff scan each garment through the configured processing stages. The manager can see the current status, assigned employee and any delays.



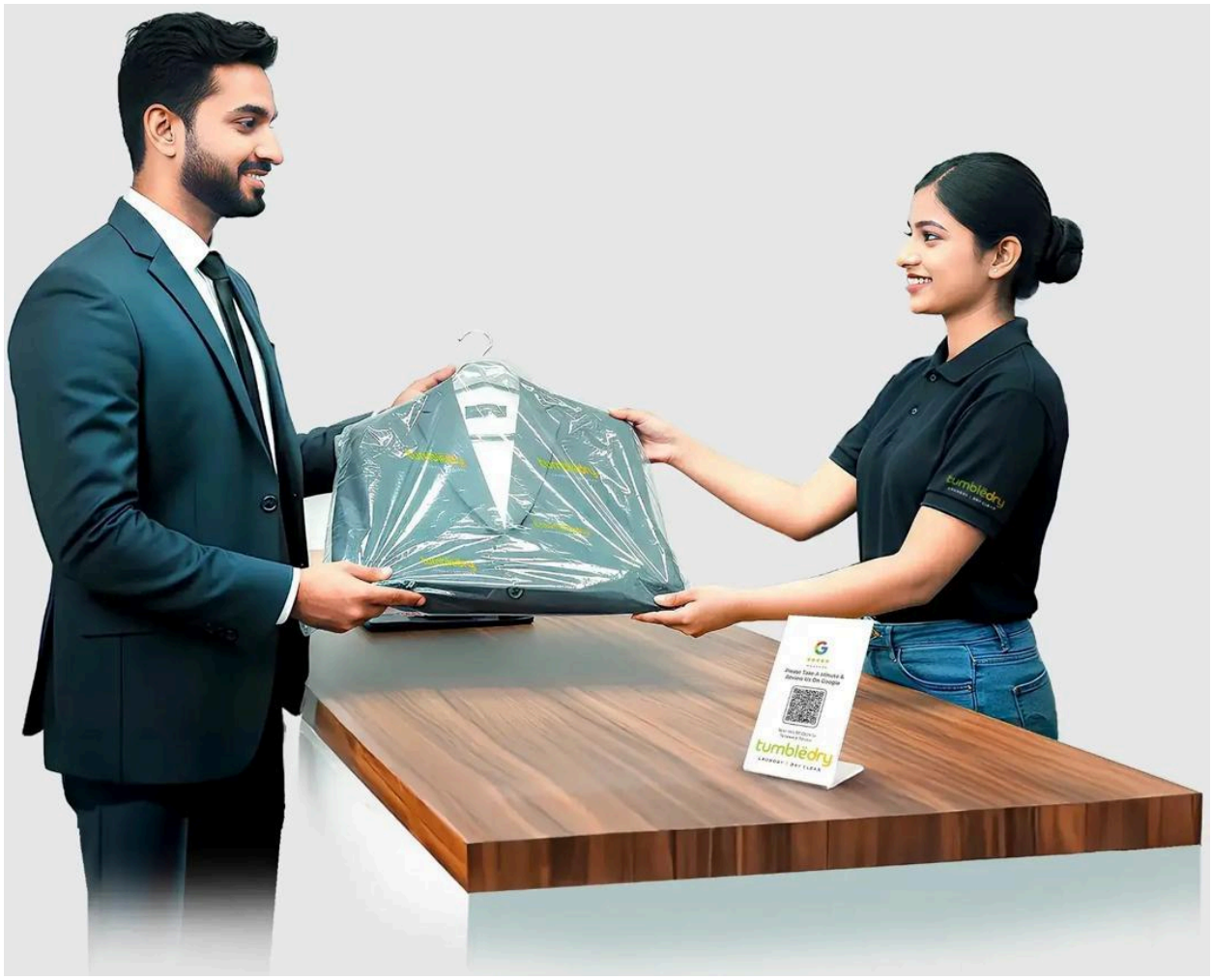
Step 5 — Quality inspection

The inspector checks each garment, records defects or rework if necessary, and approves items for packing.



Step 6 — Ready notification

The system records the garments as ready, identifies their storage location and sends the customer a collection notification.



Step 7 — Delivery and closure

The employee verifies the garments, collects the outstanding amount, issues a receipt and records the final handover. The order and payment histories remain available for future reference.

Exception scenarios the software must handle

- A garment fails inspection and returns to cleaning.
- Three garments are ready, but a fourth is delayed.
- The customer disputes pre-existing damage.
- The customer pays part of the invoice through UPI and part in cash.
- The garment cannot be located at the expected rack.
- The customer requests cancellation after work has started.
- A payment gateway reports a timeout even though the transaction may have succeeded.
- A branch transfers garments to a central cleaning plant.

Each exception needs an explicit workflow, authorized actions and an audit trail. These are not optional edge cases in a production system.

8. Functional requirements and acceptance criteria

Every feature should have a testable definition of completion. This prevents a project from being considered complete merely because screens have been built.

ID	Requirement	Acceptance criteria
FR-01	Customer management	A user can register, search, edit and view an authorized customer's history.
FR-02	Order creation	An order can contain multiple garment items with separate services and prices.
FR-03	Garment identification	Every physical garment receives a unique, scannable identifier.
FR-04	Status tracking	Every valid status change records actor, time and previous/new status.
FR-05	Pricing	Totals are calculated consistently using configured prices, discounts and taxes.
FR-06	Payment management	Partial payments, multiple payment methods and refunds reconcile correctly.
FR-07	Quality control	Garments cannot be released through the normal workflow before required checks pass.

FR-08	Delivery	The system records handover confirmation and any remaining balance.
FR-09	Notifications	Eligible events create correctly targeted messages, with delivery results and retry history.
FR-10	Reports	Figures reconcile with the underlying orders, payments, refunds and expenses.
FR-11	Access control	Unauthorized users cannot access restricted records or perform protected actions through the API.
FR-12	Data recovery	A backup can be restored and critical order and payment records verified.

Non-functional requirements

The software must also meet measurable standards for performance, reliability, security and usability.

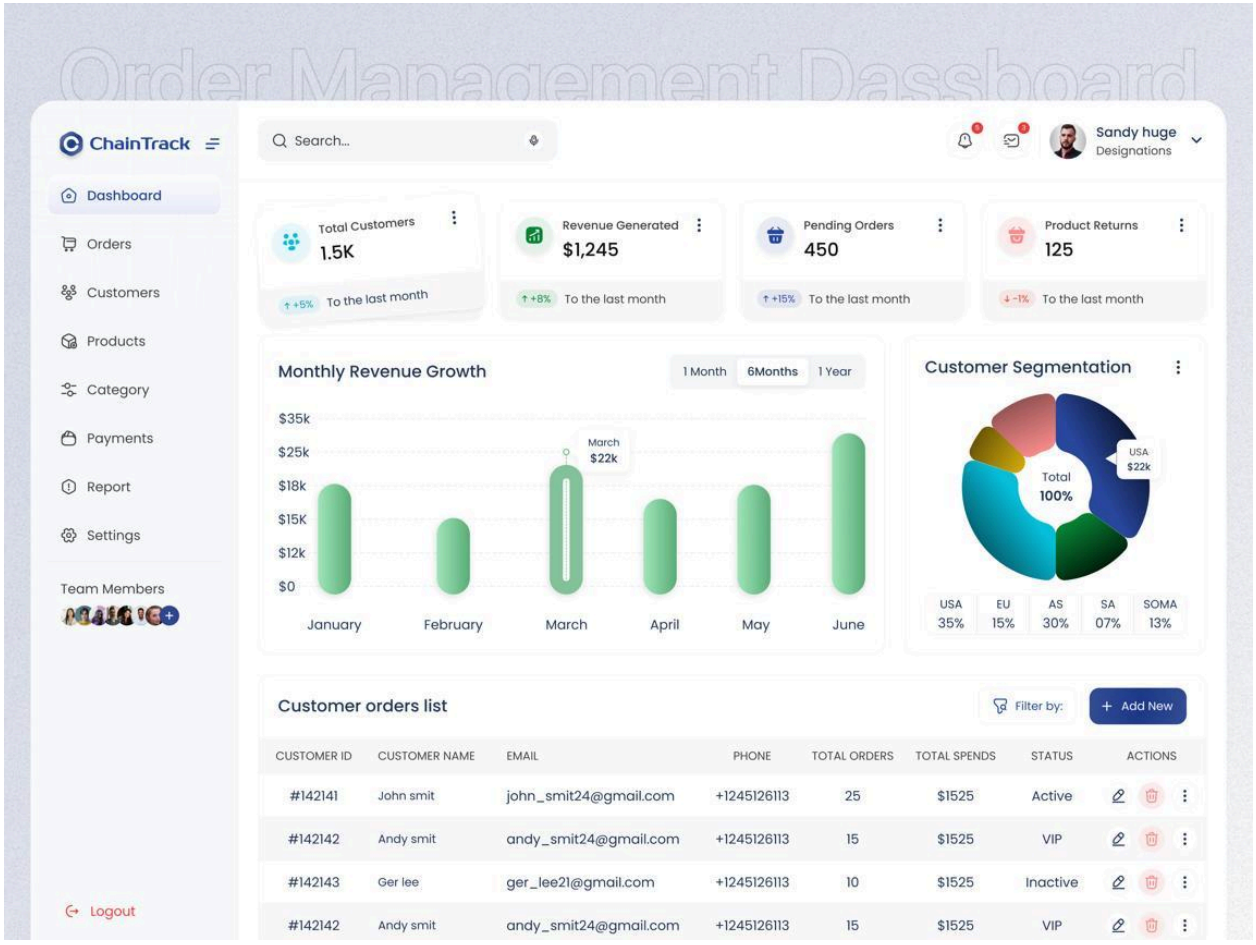
- Performance: Target a response time of under 500 ms at the 95th percentile for ordinary read APIs under the agreed pilot workload, excluding external service latency.
- Availability: Define an initial service availability target, for example 99.5% per month, with documented maintenance exclusions.
- Security: Apply server-side authorization, validated input, secure authentication and tenant isolation.
- Reliability: Prevent duplicate orders and duplicate payment processing during retries.
- Usability: Common counter operations should require as few screens and manual entries as practical.
- Auditability: Sensitive actions must be attributable to a user and timestamp.
- Recovery: Establish and test recovery-point and recovery-time objectives before production.
- Scalability: Support additional branches and businesses without redesigning the core domain model.

These are proposed engineering targets, not claims that the software already meets them.

9. User experience and screen design

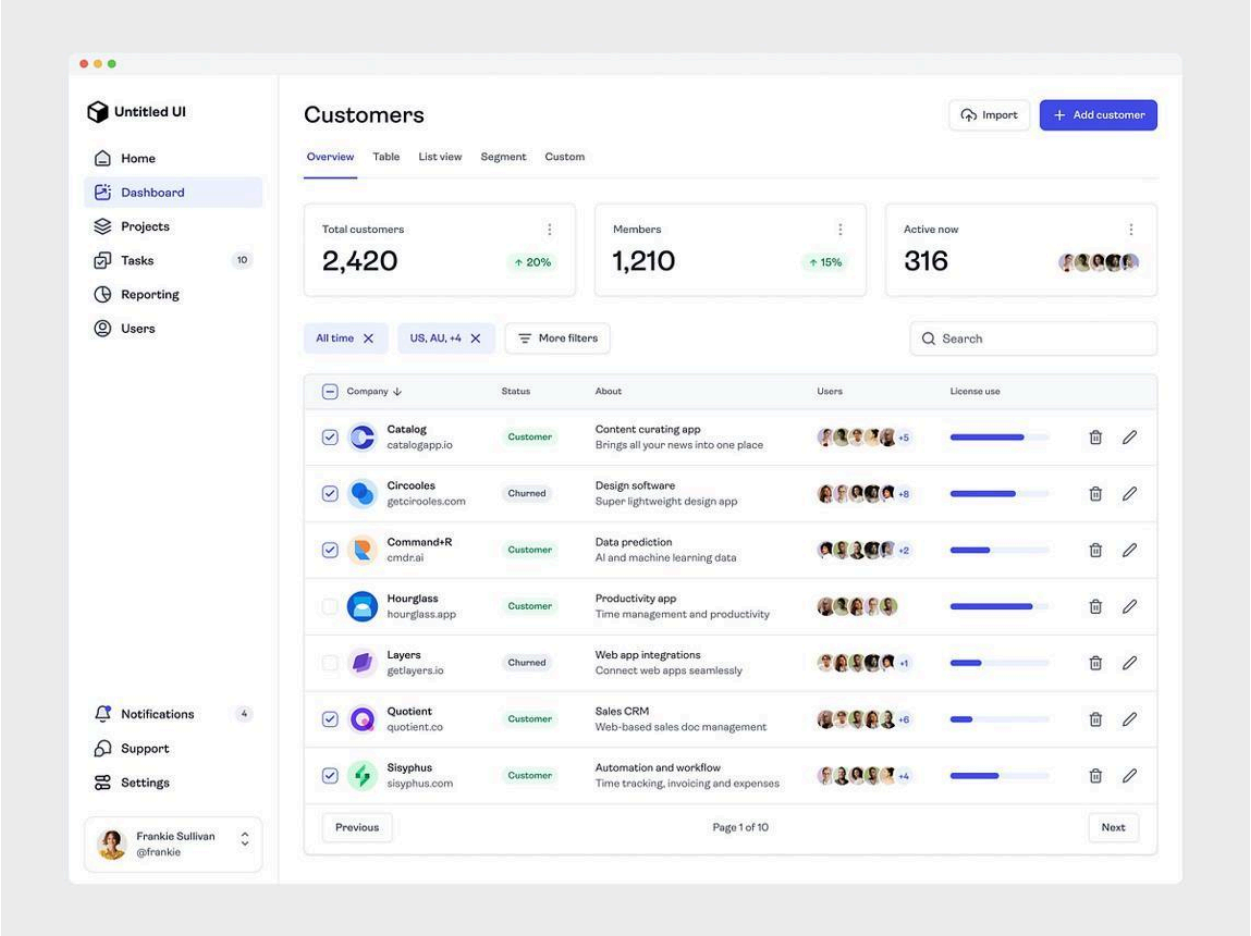
The application should be designed around actual store workflows, not around a long list of admin menus.

Primary application screens



1. Owner dashboard

KPIs, revenue, order workload, overdue orders and business alerts.

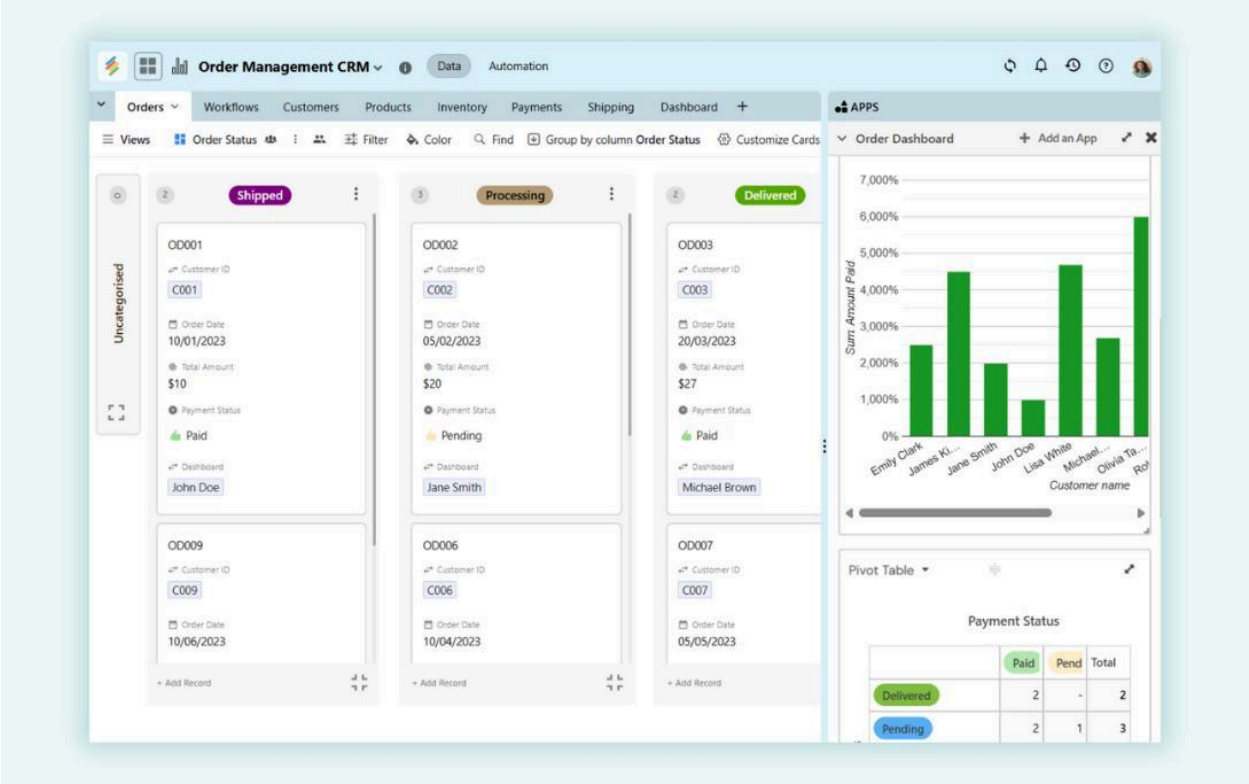


2. Customer directory

Search, customer details, history and account balances.

3. POS and order intake

Customer search, garment entry, service pricing and payment.



4. Production board

Processing stages, due dates, garment counts and rework.

Key UX principles

1. Put customer search and new order creation at the center of the counter experience.
2. Make barcode scanning the fastest way to retrieve a garment record.
3. Display order number, due date, balance and status prominently.
4. Use consistent status labels across the reception, production and delivery screens.
5. Keep the garment-level details accessible without forcing staff to open several nested pages.
6. Require confirmation for destructive actions, refunds and sensitive overrides.
7. Make tables searchable, filterable and usable on smaller screens.
8. Provide clear empty states, validation errors, loading indicators and recovery from network interruptions.

10. Technical architecture and data flow

The recommended architecture uses a modular backend, a relational database and asynchronous processing for notifications and background work.

Client applications

POS • Admin web • Production tablet • Driver mobile web • Customer portal

API layer — Fastify + TypeScript

Authentication • Validation • Authorization • Rate limiting

Business modules

CRM, orders, garment tracking, pricing, payments, inventory and reporting

Background workers

Notifications, retries, scheduled jobs and report generation

PostgreSQL

Transactional source of truth

Redis

Queues, caching and coordination

External integrations: payment gateway, messaging provider, barcode/label printer, receipt printer and optional object storage for photographs and documents.

Recommended technology choices

Layer	Proposed technology	Rationale
Frontend	Next.js, TypeScript	Responsive web application and reusable UI components
Styling	Tailwind CSS, shadcn/ui	Consistent, maintainable interface
Backend	Node.js, Fastify, TypeScript	Modular APIs and validation
ORM	Drizzle ORM	Typed schema and controlled migrations

Database	PostgreSQL	Transactions, relationships and reporting
Queue and cache	Redis with a job queue	Retryable asynchronous tasks
Authentication	Secure server-side sessions or carefully managed tokens	Controlled access and revocation
File storage	S3-compatible object storage	Garment photos, documents and generated files
Payments	Appropriate Indian payment gateway	Online payment processing and reconciliation
Messaging	Approved SMS and WhatsApp provider	Order updates and notifications
Deployment	Docker, reverse proxy, automated CI/CD	Repeatable deployment and controlled releases

Critical architectural decisions

1. One source of truth. PostgreSQL should hold authoritative order, garment, invoice and payment records. Redis must not be the only place where critical transactions are stored.

2. Separate order status from payment status. An order may be ready but unpaid, or paid but still in production. These are separate business dimensions.

3. Use controlled state transitions. The backend must validate each transition and its prerequisites rather than accepting arbitrary status strings from the frontend.

4. Use transaction boundaries. Creating an order, assigning its garments and generating initial financial records must either complete consistently or roll back.

5. Handle external integrations asynchronously. A temporary SMS or WhatsApp failure should not undo a successful order. Use a durable queue or transactional outbox, idempotent handlers and retry policies.

6. Build for tenant isolation if commercial SaaS is intended. Tenant ownership must be enforced across database access, file access, background jobs, reports and exports.

11. Database design and core business entities

The database should represent the business's real-world objects and relationships.

Organization / Tenant

Owns branches, users, customers, configuration and business records

Branches

Users and roles

Customers

Addresses • Preferences • Complaints • Account history

Orders

Order items • Garment tags • Service prices • Workflow history

Financial records

Invoices, payments, allocations, refunds

Operational records

Inspections, stock, dispatch, delivery and claims

Essential database tables

The initial schema should include:

- `organizations`, `branches`, `users`, `roles`, `permissions`
- `customers`, `customer_addresses`, `customer_preferences`
- `garment_types`, `services`, `price_lists`, `service_prices`
- `orders`, `order_items`, `garment_tags`, `order_status_history`
- `quality_inspections`, `rework_records`, `storage_locations`
- `invoices`, `invoice_items`, `payments`, `payment_allocations`, `refunds`

- `expenses`, `suppliers`, `inventory_items`, `stock_movements`
- `pickup_deliveries`, `delivery_events`, `complaints`, `claims`
- `notification_jobs`, `notification_attempts`, `audit_logs`

Data integrity requirements

- A customer can have multiple orders.
- An order can contain multiple garments.
- Every physical garment must have its own unique tag.
- One payment can be allocated across multiple invoices only if the business rules explicitly support it.
- Partial payments must not incorrectly mark an invoice as fully paid.
- Historical prices and tax breakdowns must be preserved on invoices.
- Refunds must reference the original payment or transaction being reversed.
- Status history must remain available even after an order is closed.
- Every tenant-owned record must be protected by organization-level authorization.
- Soft deletion should not remove financial or audit records that must be retained.

For reporting, use clearly defined business rules for revenue recognition, tax treatment, refunds and inventory consumption. Otherwise, two screens may display different totals for the same date range.

12. Business model and commercial opportunity

There are two possible product strategies.

Model A — Software for one business

The company owns the software and uses it for its own store or chain.

- One business's operational requirements.
- Faster deployment and simpler configuration.
- No need for customer-facing SaaS subscriptions.
- Focus on reducing manual work and improving service.

Model B — Multi-tenant SaaS

The company sells access to many independent dry cleaning businesses.

- Each business receives isolated data and configuration.
- Subscription plans, billing, onboarding and support.
- Multiple branches under one customer account.

- Platform-wide administration, usage limits and subscription lifecycle.

Illustrative SaaS pricing hypothesis

These prices are starting points for market testing, not verified market rates.

Plan	Illustrative monthly price	Proposed scope
Starter	₹799	One outlet, customer records, orders and basic billing
Professional	₹1,499	Barcode tracking, notifications, inventory and advanced reports
Business	₹2,999	Multiple branches, enhanced permissions, loyalty and delivery tools
Enterprise	Custom	Corporate accounts, integrations, onboarding and negotiated limits

Annual discounts, onboarding charges, printer hardware, messaging charges, taxes and premium integrations should be considered separately.

Example recurring revenue scenarios

Assuming the following hypothetical customer counts and an average subscription revenue of ₹1,499 per tenant per month:

Illustrative monthly recurring revenue

Revenue at different hypothetical paying-tenant counts, before costs, taxes, discounts and churn.

₹0₹200K₹400K₹600K₹800K25 tenants100 tenants250 tenants500 tenants

For example, 100 paying tenants at ₹1,499 each produce ₹1,49,900 in gross monthly subscription revenue. This is not profit. Hosting, support, sales, onboarding, payment fees, messaging and customer acquisition must be deducted.

SaaS platform features

If the product is sold commercially, add a dedicated platform administration area for:

- Tenant registration, verification and onboarding.
- Subscription plans, billing cycles, renewals and failed-payment handling.
- Trial periods, usage limits, plan upgrades and downgrades.
- Branch and user limits by plan.
- Feature entitlements and plan-specific permissions.
- Support tickets and tenant health monitoring.
- Subscription invoices and payment reconciliation.
- Tenant suspension, reactivation and data export.
- Platform audit logs and operational monitoring.

Tenant suspension must not destroy customer records. Data retention, cancellation, export and deletion policies should be clearly defined.

13. Financial feasibility for a single dry cleaning store

The software can help management understand store economics, but the financial benefits should be measured rather than assumed.

Consider this hypothetical monthly operating model:

Metric	Example
Orders per day	40
Operating days per month	26
Average order value	₹350

Monthly orders	1,040
Gross billed sales before adjustments	₹3,64,000

The calculation is:

$$40 \times 26 \times ₹350 = ₹3,64,000$$

This is illustrative gross billed sales before applicable adjustments. It is not net revenue or profit.

A useful financial model should separately track:

- Sales and applicable taxes.
- Discounts, cancellations and refunds.
- Consumables and outsourced processing costs.
- Staff costs, rent, utilities and delivery costs.
- Payment processing fees.
- Equipment maintenance and depreciation.
- Software, hosting and support expenses.
- Outstanding receivables and cash actually collected.

The owner can then determine whether software-driven improvements are translating into real economic value.

14. Implementation timeline and resource plan

A reasonable initial planning estimate for a single-store MVP is 8–12 weeks, assuming a small experienced team, clear requirements, access to the necessary hardware and integrations, and limited scope. A commercial multi-tenant product will usually need additional time for tenant administration, subscriptions, security hardening, onboarding and support workflows.

The estimates below are planning assumptions, not guaranteed delivery dates.

Phase	Estimated duration	Main deliverables

Discovery and requirements	1 week	Business workflows, user stories and acceptance criteria
UX and architecture	1 week	Wireframes, database design, API contracts and deployment plan
Foundation	1–2 weeks	Authentication, permissions, organizations, branches and configuration
Customer and POS	2 weeks	Customer CRM, orders, pricing, invoices and payments
Garment operations	1–2 weeks	Barcode tags, processing stages, inspection and packing
Integrations and reports	1–2 weeks	Notifications, printers, payment integration and core reporting
QA and pilot	1–2 weeks	Scenario testing, data migration, staff training and pilot corrections

Some phases can overlap, so these durations should not simply be added together. Hardware testing, production readiness and integration dependencies can extend the schedule.

Recommended team

- Product owner or business analyst.
- UI/UX designer.
- Frontend developer.
- Backend developer.
- QA engineer.
- DevOps support, either dedicated or part-time.
- Dry cleaning store manager or experienced operator for workflow validation.

A smaller team can combine responsibilities, but testing and real-world operational validation should not be skipped.

15. Pilot rollout strategy

The first release should be introduced in one store before expanding to every branch.

Stage 1 — Process discovery

Observe how the store accepts, tags, processes, bills and returns garments. Document existing exceptions, price lists and daily reports.

Stage 2 — Configuration and migration

Configure services, garment types, prices, users and tax rules. Import customer records and open orders only after cleaning and validating the data.

Stage 3 — Staff training

Train reception, cleaning, quality control, cashier and delivery staff using realistic sample orders and garment tags.

Stage 4 — Controlled pilot

Use the software with a limited but representative workload. Reconcile every order, garment, payment and delivery against the physical process.

Stage 5 — Evaluation

Compare baseline and pilot metrics, investigate failures, fix issues and obtain staff feedback.

Stage 6 — Expansion

Roll out to the full store, then additional branches or SaaS customers using a repeatable onboarding checklist.

Do not stop maintaining reliable financial records until the new system has been reconciled and approved for production. A temporary parallel process may be necessary during the pilot, but it should have a clear end date and a single authoritative source for each transaction.

16. Key performance indicators and success measurement

Collect baseline measurements before implementation. Then compare the same metrics over a suitable period, accounting for seasonality, staffing changes and changes in order volume.

KPI	Definition	Suggested initial target
Garment traceability	Percentage of garments with a valid tag and complete movement history	At least 98%
Order accuracy	Percentage of orders without a confirmed intake or fulfillment error	At least 99%
On-time completion	Percentage of orders ready by the promised time	Improve against baseline
Payment reconciliation	Percentage of transactions reconciled without unexplained discrepancies	100% of recorded transactions accounted for
Customer record completeness	Percentage of active profiles meeting required data rules	At least 95%
Notification delivery	Percentage of eligible notifications successfully delivered	Establish a baseline and improve
Inventory accuracy	Agreement between recorded and physically counted stock	Improve against baseline
Rework rate	Percentage of garments requiring additional processing after initial quality control	Reduce against baseline

Customer retention	Percentage of customers who return within a defined period	Improve against baseline

The numerical targets are proposed operational goals, not industry benchmarks or guaranteed outcomes. A pilot should confirm whether each goal is realistic and how it will be measured.

Measurement methodology

For each KPI, document the numerator, denominator, source table, exclusions and reporting period. For example, on-time completion should compare the recorded ready-for-collection timestamp with the promised completion time, excluding only documented and approved exceptions.

17. Risks, constraints and mitigation

Risk	Potential consequence	Mitigation
Incorrect garment tagging	Lost garments and customer disputes	Unique tags, scan verification and physical counts
Poor data migration	Duplicate profiles and incorrect balances	Validation, deduplication and reconciliation before import
Staff resistance	Incomplete records and workarounds	Simple UX, hands-on training and clear procedures
Printer or scanner issues	Slower counter operations	Test supported hardware and provide manual recovery procedures
Internet outage	Interrupted transactions	Network contingency and carefully designed offline workflows

Duplicate payment requests	Financial discrepancies	Idempotency, transaction verification and reconciliation
Unauthorized data access	Privacy and business risk	Backend authorization, tenant isolation and audit logging
Notification failures	Customers miss important updates	Retry queues, delivery logs and alternative channels
Incorrect reports	Poor business decisions	Defined accounting rules and reconciliation tests
Excessive initial scope	Cost and delivery delays	Prioritized MVP and staged release
SaaS tenant leakage	Severe cross-business data exposure	Tenant isolation tests and strict authorization boundaries
Unverified financial gains	Unrealistic business expectations	Baseline measurement and evidence-based pilot evaluation

18. Testing and quality assurance strategy

The software must be tested at several levels.

Unit testing

Test pricing calculations, discounts, taxes, payment allocation, order state transitions, permissions and inventory calculations.

Integration testing

Test the interaction between PostgreSQL transactions, APIs, background jobs, payment gateways, notifications and printer integrations.

End-to-end testing

Simulate a complete journey from customer registration to garment intake, processing, inspection, billing, collection and final reconciliation.

Security testing

Verify authentication, authorization, tenant boundaries, input validation, session management, audit logging and access to stored photographs.

Operational testing

Simulate network interruptions, repeated requests, printer failures, notification outages, concurrent edits and database restoration.

User acceptance testing

Have actual reception and production employees complete realistic workflows without developer assistance. Record friction points and fix them before full rollout.

Definition of production readiness

The system should not be released until:

- Critical customer, order and payment workflows pass.
- Financial totals reconcile with the transaction ledger.
- Garment identifiers remain unique and traceable.
- Role restrictions work through both the UI and API.
- Backups and restoration have been tested.
- Monitoring and error alerts are active.
- Store staff have been trained.
- A rollback and incident-response plan exists.

19. Future roadmap

After the core platform is stable, expand in stages.

Release 1

Core store operations

CRM, POS, garment tracking, cleaning workflows, invoices, payments, basic reports and access control.

Release 2

Customer experience and automation

Customer portal, pickup scheduling, WhatsApp notifications, loyalty, inventory and delivery management.

Release 3

Commercial SaaS

Tenant onboarding, subscription billing, plan entitlements, multi-branch management and platform administration.

Release 4

Advanced intelligence

Demand forecasting, route optimization, service profitability analytics, AI-assisted garment inspection and automated operational recommendations.

20. Final project recommendation

The proposed DryCare system should be treated as a business operations platform, not merely a billing application.

Its most important design decision is to maintain a complete relationship between the customer, the order, every individual garment, the garment's processing history, and the financial transactions. That foundation enables reliable delivery, accurate billing, customer service, reporting and future expansion.

For the first production release, prioritize:

1. Customer management and complete order history.
2. Garment-level barcode tracking.
3. Service catalogue, pricing and POS.
4. Cleaning stages, quality control and packing.
5. Invoices, partial payments, refunds and outstanding dues.
6. Notifications and collection or delivery confirmation.
7. Owner dashboard, audit trail, backup and reporting.

Add inventory, corporate accounts, memberships and SaaS subscriptions after these core workflows have been validated.

The expected business benefits—better traceability, more consistent processing, fewer manual errors and improved financial visibility—should be demonstrated through the pilot rather than assumed in advance.

Next step: turn this case study into a development specification

I recommend creating a formal Software Requirements Specification (SRS) with detailed user stories, page-by-page UI requirements, API endpoints, PostgreSQL schema, role-permission matrix, workflows, validation rules, test cases and phased implementation tasks.